



From Lysosomal Storage Disorders to Parkinson's Disease – Challenges and Opportunities

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Abstract

Lysosomes are specialized organelles with an acidic pH that act as recycling hubs for intracellular and extracellular components. They harbour numerous different hydrolytic enzymes to degrade substrates like proteins, peptides, and glycolipids. Reduced catalytic activity of lysosomal enzymes can cause the accumulation of these substrates and loss of lysosomal integrity, resulting in lysosomal dysfunction and lysosomal storage disorders (LSDs). Post-mitotic cells, such as neurons, seem to be highly sensitive to damages induced by lysosomal dysfunction, thus LSDs often manifest with neurological symptoms. Interestingly, some LSDs and Parkinson's disease (PD) share common cellular pathomechanisms, suggesting convergence of aetiology of the two disease types. This is further underlined by genetic associations of several lysosomal genes involved in LSDs with PD. The increasing number of lysosome-associated genetic risk factors for PD makes it necessary to understand functions and interactions of lysosomal proteins/enzymes both in health and disease, thereby holding the potential to identify new therapeutic targets. In this review, we highlight genetic and mechanistic interactions between the complex lysosomal network, LSDs and PD, and elaborate on methodical challenges in lysosomal research.

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Introduction – Lysosomal (dys)functions and correlations to Parkinson's disease

Parkinson's disease (PD) is a progressive neurodegenerative disorder characterized by the irreversible loss of dopaminergic neurons in the *substantia nigra pars compacta*. Worldwide, about 1% of people above 60 years are affected by PD, yet approximately 10% of PD cases manifest at younger ages (as early as patients in their 30s). These are referred to as early onset PD (EOPD).^{1,2} A large number of studies revealed a

family history in 15–25% of the diagnoses, but only 5–10% of all cases could be identified as monogenic forms, caused by a single mutation of a high risk gene.³ To date, pathogenic mutations of several genes could be identified as an inheritable source for (EO)PD, e.g., *SNCA*, *LRRK2*, *VPS35* (autosomal dominant), and *PRKN*, *PINK1*, *DJ1* (autosomal recessive).⁴ However, PD is a multifactorial disease, which appears with various symptoms and underlying pathomechanisms. Besides genetic factors, various lifestyle and environmental factors contribute to its development, and combinations thereof.⁵ Nonetheless, the growing number of genetic variants identified or proposed to contribute

to the disease implies that such genetic factors could play an important role in a great percentage of idiopathic PD cases as well.

The pathohistological hallmark of PD are Lewy bodies consisting mainly of aggregated α -synuclein (aSyn) and other aggregation-prone proteins.⁶ Interestingly, a number of recent studies show mechanistic as well as genetic connections between lysosomal (dys)function and PD pathology. This highlights the role of lysosomal pathways in the development of neurodegeneration, and proposes a complex interplay between lysosomal neurodegenerative disorders and PD caused by mutations in genes coding lysosomal proteins.

Structure and function of the lysosomal system

Lysosomes are specialized organelles, which degrade and recycle proteins, glycolipids and other intra- and extracellular components. They have an indispensable role in multiple essential processes such as autophagy and endocytosis, thus providing pathogen control and depletion of toxic substrates and degradation products.⁷ Lysosomal membranes embed ATP-consuming proton pumps to retain a luminal pH of 4.6 to 5.0.^{8,9} The acidic pH is essential for the activity of over 60 highly specialized hydrolases,¹⁰ as it assures their activity exclusively in the lysosome, thus protecting cell compartments with cytosolic pH from arbitrary degradation. It should be mentioned, however, that some lysosomal hydrolases also exhibit enzymatic activity at neutral pH.^{11–13}

The group of lysosomal hydrolases encompasses nucleases, glucosidases, phosphatases, proteases, sulfatases, and lipid or polysaccharide degrading enzymes.^{10,14,15} Lysosomal proteins are transported into the endoplasmic reticulum (ER) and travel through the Golgi apparatus where they undergo post-translational modification. To ensure targeting to the lysosome, most lysosomal proteins are tagged with mannose 6-phosphate, which is recognized by the mannose-6-phosphate receptor (M6PR). The cytoplasmic tail of the M6PR and specific sorting signals then direct the enzyme-M6PR-complex to the lysosome. While binding is most effective at pH 6.0–7.0, the acidic pH in lysosomes provokes release of the bound enzyme, allowing the M6PR to be recycled for further transport processes.¹⁶ Two forms, commonly termed cation-independent¹⁷ and cation-dependent¹⁸ M6PR, are essential players in maintenance of proper lysosomal function.¹⁹ Although most lysosomal enzymes are transported via the M6PR pathway, exceptions exist where specialized transport proteins facilitate M6PR-independent lysosomal trafficking. One of these examples is the lysosomal integral membrane protein 2 (LIMP-2), which exclusively transports β -

glucocerebrosidase (GCase) to the lysosome.^{20,21} In total, more than 50 membrane-associated proteins maintain the lysosomal environment, perform metabolic signalling and orchestrate import and export of hydrolases, substrates, metabolites and ions.¹⁰ As indicated in Figure 1, essential pathways in preserving lysosomal function and homeostasis are (i) trafficking of lysosomal enzymes/proteins via the secretory pathway, (ii) maintenance of the acidic pH, (iii) efficient turnover of substrates, (iv) export of certain substrates and metabolites (i.e. via exocytosis), and (v) communication with the cell membrane. Further, autophagy processes are essential for the functionality of the whole cell, as accumulation of toxic substrates, pathogens, cellular debris, and fragmented proteins can influence cellular and protein functions, at most even inducing cell death.²² Macroautophagy is crucial for organelle recycling/renewal and to remove aggregated proteins and cell debris from cells. Key players in this process are autophagosomes that form in the cytoplasm where they engulf their substrates before fusing with lysosomes, thus forming autolysosomes. As an alternative pathway to enter lysosomal degradation, substrates can be taken up via chaperone-mediated autophagy.²³ For proper autophagic function, a tight regulation is indispensable, including up- or downregulation of proteins, and signal cascades for autophagosome or autolysosome formation.^{23,24} Important components are lysosome-associated Hsc70 chaperones that bind to substrates and transport them to the lysosome-associated membrane protein 2 (LAMP2) to form a translocation complex that delivers the substrate into the lysosomal compartment.²⁵ Several other proteins, e.g., transcription factor TFEB, and leucine-rich repeat kinase 2 (LRRK2), affect autophagic function of cells.^{26–28}

Failures in the function of lysosomes can lead to various diseases, most notable to those referred as lysosomal storage diseases or disorders (LSDs). At least 50 types of LSDs have been described to date that are mostly of autosomal-recessive character.^{29,30} They are characterized by defects of lysosomal proteins and consequent accumulation of undegraded macromolecules and are largely caused by genetic variations in genes coding for lysosomal hydrolases. LSDs often manifest with neurological symptoms and neurodegeneration³¹ due to accumulating substrates causing cellular dysfunction. Neurons are especially sensitive towards the accumulation of cellular waste, in part because of their post-mitotic character. Consequently, substrate accumulation in the brain often leads to neurotoxicity and irreversible cell loss.^{32,33} Furthermore, insufficient lysosomal degradation includes scant removal of neurodegeneration-associated proteins like the aggregation-prone aSyn (encoded by *SNCA*) or other harmful molecules.^{24,34,35} As a result, accumulation of undegraded substrates can serve as a trigger in ini-

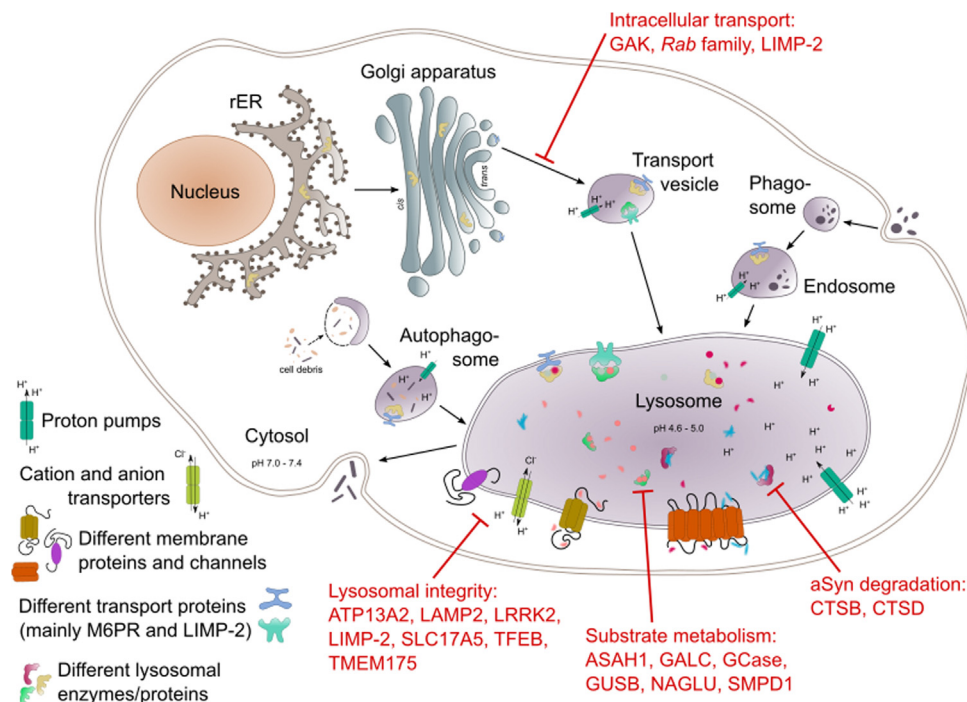


Figure 1. Overview of lysosomal formation and function highlighting corresponding PD-associated pathways and selected genetic variants. Lysosomes are acidic vesicles with a lipid-bilayer membrane. They consist of specialized lysosomal membrane proteins and hydrolases that maintain lysosomal homeostasis. Relevant steps in lysosome formation and maintenance are indicated together with examples of lysosomal proteins of which certain genetic variants have been identified as PD risk factors. This includes lysosomal membrane proteins affecting lysosomal integrity, lysosomal enzymes for the degradation of various substrates including the PD-associated protein α -synuclein (aSyn), as well as components of the intracellular trafficking machinery.

tiating secondary pathogenic cascades, affecting molecules not directly related to the defective enzyme. This causes, for example, perturbation in membrane composition, impaired intracellular vesicle fusion and trafficking, impaired autophagic function, disturbance in calcium homeostasis, mitochondrial abnormalities and excessive accumulation of reactive oxygen species.³⁶

LSDs often share symptoms with PD and different forms of parkinsonism and a growing number of studies showed connections on the genetic and the molecular level between the two diseases, as summarized in Table 1. This raises the question whether these disorders share common or similar underlying pathomechanisms. In the following chapters, we will present examples of lysosomal dysfunctions to illustrate how the understanding of lysosomal (dys)functions can lead to a better comprehension of PD pathology and the identification of novel biomarkers, diagnostic tools or therapeutic targets.

α -Synuclein degradation by lysosomal cathepsins

A direct link between lysosomal function and PD might be the degradation of aSyn within the lysosomal system. While the proteasome is also

involved in aSyn degradation³⁷, clearance of aSyn aggregates is preferentially mediated through endocytosis, macroautophagy and chaperone-mediated autophagy^{38–40}. The bulk degradation of protein substrates, including aggregated aSyn, within the lysosome is mainly mediated by lysosomal cathepsins. These cathepsins do not only degrade proteins, but are also crucial for processing, maturation and regulation of other lysosomal proteins and enzymes. Thus, it is unsurprising that deficiency of lysosomal cathepsins can disrupt lysosomal function and lead to LSDs. Examples are neuronal ceroid lipofuscinosis type 10 (NCL10) and 13 (NCL13), in which lipofuscin is insufficiently degraded by cathepsins D (CTSD) and F (CTSF), respectively^{41–43}. Another example is galactosialidosis, where the lack of cathepsin A causes dysregulation of neuraminidase-1 and beta-galactosidase, leading to accumulation of sialylated substrates.^{44,45} CTSD merits special attention here, as genetic alterations in the *CTSD* gene have not only been linked to NCL10, but also to neurodegenerative disease, namely Alzheimer disease (AD),^{46–49} and PD. Not only CTSD, but also cathepsin B (CTSB) emerged as an active player in PD pathology as both have been shown to be important for aSyn cleavage and genetic variants have been identified as risk factors for PD.^{50–53} The link

Table 1 Overview of lysosomal and lysosome-associated proteins correlated to LSDs and PD.

α-Synuclein degradation by lysosomal cathepsins					
Gene name	Protein name(s)	Functions	Consequences of dysfunction	Disease name	REF
<i>CTSB</i>	Cathepsin B	Cysteine protease; co-factor for other proteases' activity (e.g. Cathepsin D); autophagy	Reduced activity of several proteases; associated with brain ageing	AD, PD	52,282[295]
<i>CTSD</i>	Cathepsin D	Aspartate protease, degradation of long-lived proteins incl. growth factors and aSyn	Accumulation of toxic protein aggregates	Congenital neuronal ceroid lipofuscinosis type 10 (NCL10), AD, PD	41,51,282,283
<i>CTSL</i>	Cathepsin L	Cysteine protease; association with the activation of NF-kappa B; autophagy; protein degradation	Upregulated by mitochondrial stress, leading to altered distribution and increased autophagy and dopaminergic neuron death; increased levels in PD		284
<i>LGMN</i>	Legumain, asparagine endopeptidase (AEP)	Cysteine protease; cleavage of tau proteins and amyloid precursor proteins; part of immune system and programmed cell death	Upregulated due to ageing and in AD		285,286
Dysfunctional substrate cleaving enzymes and impaired metabolism					
Gene name	Protein name(s)	Functions	Consequences of dysfunction	Disease name	REF
<i>ARSA</i>	Arylsulfatase A, cerebroside-sulfatase	Degradation of cerebroside 3-sulfate; activated by saposin B	Accumulation of sulfatides	Metachromatic leukodystrophy	287
<i>ARSB</i>	Arylsulfatase B	Degradation of dermatan sulfate and chondroitin sulfate	Accumulation of glycosaminoglycans	Maroteaux-Lamy syndrome	288,289
<i>ASAH1</i>	Acid ceramidase (ACDase), N-acylsphingosine amidohydrolase 1	Degradation (and build-up) of ceramides	Accumulation of ceramides causing curvilinear tubular structures	Farber disease, AD, PD	51,290
<i>FUCA1</i>	α -L-fucosidase	Hydrolysis of α -L-fucoside	Accumulation of fucoside and glycan structures	Fucosidosis, chronic autoimmune and inflammation disorders in children	291,292
<i>GALC</i>	Galactocerebro-sidase, galactosylceramidase	Hydrolysis of galactosylceramide, psychosine and monogalactosyldiglycerid	Accumulation of galactosylceramide; and psychosine, resulting in demyelination due to apoptosis of myelin-forming cells	Krabbe disease, PD	51,52,89,92,293
<i>GALNS</i>	N-acetylgalactosamine-6-sulfatase	Hydrolysis of chondroitin sulfate and keratan sulfate	Accumulation of glycosaminoglycan and glycan structures	Morquio syndrome	294
<i>GBA1</i>	β -Glucocerebrosidase (GCase)	Hydrolysis of glucosylceramide	Accumulation of glucocerebroside	GD, PD	70,75,295,296
<i>GLA</i>	α -galactosidase A (α -Gal A)	Hydrolysis of glycolipids and glycoproteins	Accumulation of glycosphingolipids	Fabry disease	297,298
<i>GNS</i>	N-acetylglucosamine-6-sulfatase	Hydrolysis of keratan sulfate and heparin/heparan sulfate	Accumulation of heparan and keratan sulphate	Sanfilippo syndrome (MPS IIID)	299

Table 1 (continued)

α-Synuclein degradation by lysosomal cathepsins					
Gene name	Protein name(s)	Functions	Consequences of dysfunction	Disease name	REF
<i>GUSB</i>	β -glucuronidase	Hydrolysis of chondroitin sulfate, dermatan sulfate, and heparan sulfate	Accumulation of glycosaminoglycans	Sly syndrome (MPS VII), PD	93,300
<i>HEXA</i>	β -Hexosaminidase subunit α	Hydrolysis of N-acetyl-D-hexosamines and G _{M2} gangliosides with subunit β	Accumulation of G _{M2} gangliosides; altered levels in PD	Tay-Sachs disease	116,119,301
<i>HEXB</i>	β -Hexosaminidase subunit β	Hydrolysis of N-acetyl-D-hexosamines and G _{M2} gangliosides with subunit α	Accumulation of G _{M2} gangliosides; altered levels in PD	Sandhoff disease	116,119,302
<i>IDS</i>	Iduronate 2-sulfatase	Degradation of heparan sulfate and dermatan sulfate	Accumulation of glycosaminoglycans	Hunter syndrome	303,304
<i>IDUA</i>	Glycosaminoglycan α -L-iduronohydrolase, iduronidase	Hydrolysis of α -L-iduronosidic linkages in glycosaminoglycans dermatan and heparan sulfate	Accumulation of glycosaminoglycans	Hurler syndrome, Scheie syndrome	305
<i>NAGA</i>	α -N-acetylgalactosaminidase, nagalase	Glycoside hydrolase; deglycosylation of Gc proteins	Excretion of glycoproteins	Schindler diseases type I and II, Kanzaki disease	306–307
<i>NAGLU</i>	α -N-acetylglucosaminidase	Hydrolysis of N-acetyl-D-glucosamine and UDP-N-acetylglucosamine	Accumulation of glycosaminoglycans	Sanfilippo syndrome (MPS IIIB), PD	308,309
<i>NEU1</i>	Neuraminidase 1, sialidase-1	Cleavage of sialic acid in glycoproteins and glycolipids; heterotrimeric complex with β -galactosidase (β -gal) and cathepsin A	Accumulation of glycosaminoglycans	Glycoproteinosis, Sialidosis (ML I)	310
<i>SGSH</i>	N-sulphoglucosamine sulphohydrolase, sulfamidase	Hydrolysis of N-sulfo-D-glucosamine	Accumulation of heparan sulfate	Sanfilippo syndrome (MPS IIIA)	308
<i>SMPD1</i>	Sphingomyelin phosphodiesterase1, acid sphingomyelinase (ASMase)	Hydrolysis of sphingomyelin	Accumulation of sphingomyelin causing lipid abnormalities in cell membranes	SMPD1-associated Niemann-Pick disease types A and B, p.L320P mutation in PD	51,107,311
Lysosomal integrity and transport equilibrium					
Gene name	Protein name(s)	Functions	Consequences of dysfunction	Disease name	REF
<i>ATP6V0C</i>	C subunit of the V0 domain of V-type proton ATPase 16 kDa	Proteolipid subunit in a vacuolar proton pump; acidification of lysosomes and endosomes	Inhibition of acidification, inhibition of autophagy processes, accumulation of aSyn, stress-induced cell death; neurodegenerative disorders		312
<i>ATP13A2</i>	Probable cation-transporting ATPase 13A2, PARK9	Export of zinc and manganese for cell export or lysosomal sequestration	Accumulation of toxic metal ions, aSyn aggregation	Kufor-Rabek syndrome, PD	140,141,313

(continued on next page)

Table 1 (continued)

α-Synuclein degradation by lysosomal cathepsins					
Gene name	Protein name(s)	Functions	Consequences of dysfunction	Disease name	REF
<i>GRN</i>	Progranulin (PGRN)	Lysosomal glycoprotein and growth factor in close interaction with cathepsins, GCase, TFEB, and many others	Reduced autophagosomal processes; decreased activity of lysosomal enzymes; saposin reduction	Batten disease or neuronal ceroid lipofuscinosis (NCL), frontotemporal lobar degeneration with frontotemporal dementia and parkinsonism, AD	314–317
<i>LAMP2</i>	Lysosome-associated membrane protein 2	Integral membrane protein; interactions and imports substrates from other enzymes into the lysosome; essential for autophagy processes; requires Hsc70 for regeneration	Altered autophagy and chaperone mediated autophagy; accumulation of aSyn	Danon disease, PD	126,129
<i>LRKK2</i>	Leucine-rich repeat kinase 2, dardarin, PARK8	Kinase in the cytoplasm, interacting with lysosomal and mitochondrial membranes	Altered phosphorylation and protein–protein interactions, e.g. causing pH dysregulation; enhanced lysosomal localization and sorting, e.g. more autophagy processes	PD	318–321
<i>MCOLN1</i>	Mucopolipin-1, TRPML1	Transient receptor; potential cation channel; iron ion transporter	Iron deficiency in cells/lysosomes; reduces aSyn when activated	Mucopolipidosis type IV	322,323
<i>NPC1/2</i>	Niemann-Pick C1 protein (<i>NPC1</i>), epididymal secretory protein E1 (<i>NPC2</i>)	LDL-derived cholesterol exporter in lysosomal membranes	Reduced expression of myelin gene regulatory factor	Niemann-Pick disease, type C	324–326
<i>SLC17A5</i>	H(+)/sialic acid cotransporter, sialin	Sialic acid transporter in lysosomal membranes	Accumulation of sialic acid in the lysosome/cell	Salla disease, Infantile Sialic Acid Storage Disease (ISSD), PD	51,327,328
<i>TFEB</i>	Transcription factor EB	Expression regulation of lysosomal hydrolases, membrane proteins and autophagy-related proteins, activated by PGC1- α	Expression of inflammatory cytokines; accumulation of proteins (aggregates)	Huntington, PD, AD	329–331
<i>TMEM175</i>	Transmembrane protein 175	Lysosomal, proton-activated potassium channel	Altered pH in lysosomes; aSyn aggregation	Rapid eye movement sleep behaviour disorder, PD	52,327
<i>TMEM76 or HGSNAT</i>	Transmembrane protein 76, heparan- α -glucosaminide N-acetyltransferase	Lysosomal membrane protein acetylating heparin and heparan sulfate, uses acetyl coenzyme A	Accumulation of heparin or heparan sulfate	Sanfilippo syndrome (MPS IIIC)	332
Vesicular transport of lysosomal enzymes and lysosomal segregation					
Gene name	Protein name(s)	Functions	Consequences of dysfunction	Disease name	REF
<i>GAK</i>	Cyclin G-associated kinase, auxilin 2	Serine/threonine kinase; uncoating clathrin in neurons	Impaired interaction with cathepsins and aSyn accumulation; reduced autophagy	PD	333
<i>M6PR</i>	Cation-dependent mannose-6-phosphate receptor (CD-MPR)	Transmembrane glycoprotein binding mannose-6-phosphate (M6P) tags on acid hydrolase	Insufficient transport of proteins to the lysosome leading to inefficient sequestering of (toxic) cell structures		334

Table 1 (continued)

α -Synuclein degradation by lysosomal cathepsins			
Gene name	Protein name(s)	Functions	Consequences of dysfunction
<i>LMBRD1</i>	Limp region 1	precursors in the Golgi apparatus to translocate them to the lysosome Lysosomal membrane protein; complex formation and translocation of ABCD4 (cobalamin exporter); regulation of insulin receptor (endocytosis)	Accumulation of cobalamin, lack of homocysteine and methionine
<i>Rab</i> family members	Ras-related proteins, small G-proteins	Peripheral membrane proteins; vesicle transport and fusion	Cobalamin F-type disease 156,157,159
<i>SCARB2</i>	Lysosomal Integral Membrane Protein type-2 (LIMP-2)	Type III transmembrane glycoprotein; binding of GCcase in ER and transport to lysosome; catalysing GCcase activity	Griscelli syndrome type 2 (Rab27A protein), hereditary sensory neuropathy type 1C (Rab7A protein), EOPD (Rab39B protein) Action myoclonus-renal failure syndrome, GD, PD 164,165,335 20,336

Legend: Overview of proteins in lysosomes or lysosomal membranes, correlated to LSDs, PD, or other neurodegenerative disorders. Some proteins are known under more names than listed and not all associated diseases, e.g. different types of cancer, are mentioned here. The enzymes are sub-divided by their cellular localization and function as mentioned in the text, yet not all proteins of this table are explained in detail in this review. Genes are listed by alphabetic order.

between CTSD and PD is further solidified by studies of a CTSD-deficient (*Ctsd*^{-/-}) NCL10 mouse model, which displays disturbed autophagosomal function and aSyn aggregation⁵⁴. Although no genetic variants of cathepsins L (CTSL) and K have been associated with PD so far, both enzymes have been shown to cleave aSyn, even its aggregated forms.^{50,55}

Accumulation of aSyn aggregates can occur directly within the lysosome, manifesting again the organelle's key role in PD pathology⁵⁶. In line with this, up- or down-regulation of lysosomal cathepsins as a potential mechanism or even biomarker of PD has recently gained interest. However, data about CTSD levels and activity in the context of PD is ambiguous between studies, most likely due to the analysis of different body fluids and tissue samples. For example, in studies analysing blood serum samples of PD, CTSD levels were either reduced⁵⁷ or unchanged⁵⁸. On the other hand, a study looking at cerebrospinal fluid (CSF) reported CTSD activity to be unchanged in PD patients⁵⁹, while another publication reported unchanged protein levels of CTSD and CTSL but increased levels of CTSB and CTSF in PD-derived CSF samples⁶⁰. A study analysing brain tissue showed no significant changes in CTSB, D and L activity in PD patients.⁶¹ On the contrary, studies using patient-derived cell models reported increased CTSD levels in multiple Gaucher's disease (GD) and GBA-PD fibroblasts harbouring different mutations (p.N370S, p.L444P, c1263del55, 203insC, RecNcil)⁶² and iPSC-derived neurons of different p.N370S mutation-carrying GBA-PD patients.⁶³ Characterization of selected PD-associated CTSD variants (p.V95I, p.G145V, p.A239V, p.R266H, p.D97S) reported no significant impact of these mutations on enzyme activity, except for p.A239V, which increased activity in an *in vitro* study.⁴³ Thus, the exact role and potential as biomarker of lysosomal cathepsins in PD remains a topic of discussion. It is important to mention that protein levels and activity of lysosomal hydrolases are not equivalent read-outs because impairments in cellular transport or maturation may not affect total cell protein levels, but will affect lysosomal activity. Nonetheless, modulation of lysosomal cathepsin activity may be a powerful tool in the treatment of PD as shown by a recent study which explored administration of recombinant CTSD as a therapeutic approach to reduce aSyn burden.⁶⁴ This study showed that an administered recombinant pro-form of CTSD was taken up and processed into catalytically active enzyme by iPSC-derived neurons. Furthermore, treatment with recombinant CTSD reduced aggregated aSyn levels and restored overall lysosomal function in PD patient-derived dopaminergic neurons harbouring the *SNCA* p.A53T mutation.⁶⁴ Importantly, the positive effects of CTSD treatment on aSyn clearance were reproduced in the already mentioned CTSD-deficient (*Ctsd*^{-/-}) NCL10 mouse model, in

which previous studies had already shown that intracranial injection of these mice with recombinant CTSD improved autophagy, life span, and disease phenotype.⁶⁵

In summary, lysosomal cathepsins appear to play a central role in the maintenance of lysosomal function and the degradation of aSyn. However, due to the heterogeneous data on dysregulation of different cathepsins in PD along with the complexity of regulatory interactions within the protease web of the lysosome, we refrain from providing a general conclusion about the exact mechanisms by which different cathepsins modulate PD pathology. Nonetheless, therapeutic intervention targeting cathepsin activity, as shown for CTSD, could be a promising approach towards tackling neurodegeneration.

Dysfunctional substrate cleaving enzymes and impaired metabolism

As the brain is characterized by a high lipid content with dense lipid rafts in the neuronal membrane,⁶⁶ ceramides – fundamental components of all major sphingolipid species – play an important role in the regulation of central nervous system (CNS) homeostasis.⁶⁷ Glucosylceramide and galactosylceramide are sphingolipids degraded within the lysosome by GCase and β -galactosylceramidase (GALC), respectively. For both enzymes, several hundred mutations have been identified on the genomic level. Loss of function mutations in *GBA1* (encoding GCase) and *GALC* genes cause the LSDs GD^{68–70} and Krabbe disease,^{71,72} respectively. For GD, three different types are defined based on the severity of symptoms.⁷³ While patients suffering from type 1 show no neurological symptoms, type 2 and 3 patients develop neurological problems. Here, type 2 is most severe and patients normally report with symptoms as infants and, if untreated, die around the age of 9 months.⁷⁴ Interestingly, an increased incidence of *GBA1* mutations was found in large cohorts of PD patients identifying *GBA1* as risk factors for PD.⁷⁵ Further, neuropathological analyses of GD patients revealed the presence of aSyn-positive aggregates underlining the contribution of aSyn in GD pathology.³⁴⁰ While manifestation of GD requires homozygous or compound heterozygous *GBA1* mutations, specific heterozygous *GBA1* genotypes like the p.N370S mutation are sufficient for an increased PD risk. On the molecular level it was demonstrated that the substrate of GCase, glucosylceramide, interacts with and stabilizes aSyn oligomers inducing the formation of cell toxic aSyn species.^{76,270} Knockdown and inhibition of GCase *in vitro* and *in vivo* resulted in impaired lysosomal clearance capacity and aSyn accumulation.²⁷⁰ Additionally, *in vitro* findings showed that glucosylceramide stabilizes the soluble high molecular weight intermediate

forms of the protein.^{76,270} Accumulation of these toxic oligomeric species not only leads to neurodegeneration, but also causes the depletion of lysosomal GCase. This bi-directional effect of GCase and aSyn creates a vicious circle, which can lead to a self-propagating disease.⁷⁶ Furthermore, a feedback loop between aggregated aSyn caused reduction of proteinphosphatase 2A activation, leading to a lack of aSyn dephosphorylation, has been proposed.⁷⁷ As ceramide acts as an activator of proteinphosphatase 2A, accumulation of aggregation prone phosphorylated aSyn species can occur also due to insufficient proteinphosphatase 2A activity.⁷⁸ Additionally, ageing is a major risk factor of neurodegenerative diseases (reviewed in⁷⁹) and ceramides have been shown to accumulate in relation with ageing⁸⁰ while being implicated in processes contributing to the pathomechanism of neurodegenerative disorders. These include cellular senescence, cell cycle arrest, and apoptosis,^{81–83} thus it is not surprising that imbalances of ceramide metabolism have been implicated in PD and other age-related neurodegenerative diseases.⁸⁴

Krabbe disease (also known as globoid cell leukodystrophy; GLD), manifests in 95% of the cases as an early form at ages between one and seven months, leading to death within the first two years.⁸⁵ Most remaining cases have onset ages between six months and three years, also resulting in an early death after two to three more years.⁸⁶ Typical hallmarks in all cases are stiffness, decreased psychomotor functions, blindness, and hyperirritability. At least 60 out of the 260 known variants of *GALC* cause different variants of Krabbe disease. Impaired enzyme activity causes accumulation of galactosylceramide in oligodendrocytes and subsequent demyelination.^{72,87,88} A side effect of the disease is the emergence of neuronal inclusions containing ubiquitin and aggregated aSyn. Analysis of *post mortem* brain tissue revealed significantly increased psychosine content in the cortex and a trend towards decrease enzyme activity in both grey and white matter of PD patients compared to controls. However, risk variants with increased enzyme activity also exist, and are associated with PD.⁸⁹ For some PD-correlated variants, psychosine concentrations were particularly increased in cortical samples of PD patients.⁹⁰ In an *in vitro* assay, psychosine was found to promote aSyn fibrillization.⁹¹ These findings support the notion that dysfunctional psychosine metabolism could lead to the accumulation of the lipid in the ageing brain, thus contributing to the risk of the development of aSyn pathology.⁹⁰ Furthermore, genome-wide association studies (GWAS) and genome sequencing in cohort studies proposed various *GALC* variants to serve as risk factors of PD.^{52,92,93} Such variant is the SNP rs8005172, which was found to be associated with PD in the Chinese population.⁹⁴

Galactosylceramide can be formed by the cleavage of sulfate groups of sulfatides, catalysed by Arylsulfatase A (ARSA), which is then further hydrolysed by GALC. Mutations within the *ARSA* gene cause impaired activity of the enzyme and lead to the accumulation of sulfatides in the nervous system and visceral organs, resulting in metachromatic leukodystrophy (MLD), a rare autosomal recessive LSD.^{95–97} Suzuki *et al.*⁹⁸ reported a possible mechanistic link between PD and MLD. They observed diffuse aSyn accumulation among axons and glial cells in brain stem and cerebral white matter samples of MLD patients. Partial loss of ARSA enzymatic activity was significantly more frequent among patients with parkinsonism and/or essential tremor compared to control individuals.⁹⁹ Later, partial ARSA deficiency was associated with motor symptoms that are also characteristic of PD, such as tremor, muscle rigidity, and bradykinesia. Thus, the authors proposed that while decreased enzymatic activity is not the primary cause of parkinsonism, it could trigger the development of symptoms when combined with other genetic and/or environmental risk factors.¹⁰⁰ With the use of cell- and nematode models, ARSA deficiency was shown to increase aggregation, secretion, and cell-to-cell propagation of aSyn.¹⁰¹ On the contrary, in an *in vitro* aSyn aggregation assay, the presence of ARSA inhibited aSyn fibrilization in a dose-dependent manner.¹⁰¹ Separate analysis of cytosolic and vesicular fractions of SH-SY5Y cells revealed that despite ARSA being a lysosomal enzyme, it is (although inactive) present in the cytosol as well.¹⁰¹ Based on results of co-immunoprecipitation experiments, cytosolic ARSA was found to interact with aSyn. The aSyn binding affinity of the putative protective p.N325S variant of the enzyme was high, whereas the p.L300S variant - resulting in total loss of enzymatic function - showed lower aSyn binding affinity compared to the wild type protein.¹⁰¹ Ectopic overexpression of the high, but not the low aSyn binding ARSA protein form rescued locomotor behaviour deficits in a fly model of the disease, and was capable of decreasing aSyn aggregation and release from ARSA knockout cells.¹⁰¹ These results suggest that ARSA is not only an important enzyme in normal lysosomal function, but also serves as a molecular chaperone of aSyn. Thus, deficiency of the enzyme could contribute to the development of PD pathology not only by causing lysosomal dysfunction, but also through enhanced aSyn aggregation.¹⁰¹ Recently, the importance of ARSA variants, specifically the protective effect of the p.N325S mutation in PD has been challenged.^{102,103} Thus, studies involving large sample size and functional analyses are highly warranted to clarify the role of ARSA in PD.

Another example for mutations in enzymes upstream of the ceramide degradation pathway are variants of acid sphingomyelinase (aSMase), an enzyme which catalyses the conversion of

sphingomyelin to ceramide. It is encoded by the Sphingomyelin Phosphodiesterase 1 (*SMPD1*) gene, in which mutations cause the LSD Niemann-Pick disease type A and B, characterized by the accumulation of sphingomyelin in the lysosome.¹⁰⁴ A link between aSMase and PD was supported when certain *SMPD1* gene variants were found in PD and control cohorts at different frequencies.^{51,105–108} In their recent study, Alcalay *et al.* assessed the frequencies of *SMPD1* variants in three different study cohorts and reported association of the p.L302P and 996delC (fsP330) variants with PD among patients of Ashkenazi-Jewish origin.¹⁰⁹ Furthermore, reduction in enzyme activity was found to be associated with earlier disease onset. *In vitro* findings demonstrated that knockout or knockdown of *SMPD1* led to an increase in aSyn levels.¹⁰⁹ Functional analysis of how these gene variants could affect aSMase revealed that both the p.L302P and fsP330 variants of the enzyme caused impaired transport of the protein to the lysosome. Furthermore, *in silico* structural analysis revealed that the L302P variant is likely to cause the catalytic domain of the enzyme to unfold.¹⁰⁹

However, not all LSDs can be linked to PD pathology. For some, a connection is controversially discussed, as seen for example in β -Hexosaminidase (β -HEX) deficiency. It is a member of the glycosyl hydrolase 20 protein family and, together with the cofactor GM₂ ganglioside activator protein, catalyses the degradation of terminal N-acetyl hexosamine-containing molecules such as the ganglioside GM₂. Mutations in the two subunits of β -HEX (subunit α and β) lead to the accumulation of GM₂, resulting in GM₂-gangliosidosis Tay-Sachs (caused by variants of the *HEXA* gene, encoding the α subunit of β -HEX) and Sandhoff disease (caused by mutations of the *HEXB* gene, encoding the β subunit). Presence of parkinsonian-like symptoms have been described in patients suffering from late/adult-onset GM₂-gangliosidosis.^{110–113} Furthermore, accumulation of aSyn in both *in vivo* models as well as *post mortem* brain samples of GM₂-gangliosidosis patients have been repeatedly reported.^{98,114,115} Elevated β -HEX activity has been reported in CSF and fibroblast samples of PD patients^{116,117}, yet not in leukocytes.¹¹⁸ More recently, a decrease in β -HEX activity was described in *substantia nigra* samples of sporadic PD patients when compared to samples of healthy controls.¹¹⁹ Considering these controversial results of β -HEX activity, a clear connection between GM₂-gangliosidosis and PD cannot be drawn.

Lysosomal integrity and transport equilibrium

Lysosomal proteins are often not directly linked to one single disease, but impose risk factors for PD, dementia with Lewy bodies, and other

neurodegenerative dysfunctions as well.³¹ The assembly, structure, and integrity of lysosomes is based on various (membrane) proteins, which ensure import and export of substrates, communication between lysosomes and other cell organelles or the cell membrane, signal transduction, pH stability, and many others.¹²⁰ Disturbed integrity can influence the entire lysosomal system and provoke aSyn aggregation or cell death, e.g., by insufficient export of substrates or degraded products, which causes accumulation of toxic elements in the lysosomes.¹²¹

Lysosome-associated membrane proteins (LAMPs) with their subtypes 1, 2, and 3, are the most common membrane proteins correlated to lysosomal integrity and segregation processes. LAMP-1 and -2 represent 50% of all lysosomal membrane proteins.¹²² Although their functions are not entirely clear to date, data obtained with the use of mouse models suggest not only a role in maintaining the lysosomal structure, but also autophagic degradation.²⁵ Especially interaction and binding of enzyme-bound substrates is often mediated by LAMP2 crosslinked with the lysosome-associated Hsc70 heat-shock protein, thereby forming large multimers to transport substrates into the lysosome. Furthermore, it has high relevance for different autophagy processes.^{25,123} LAMP3 is expressed exclusively in dendritic cells and it takes part in processes such as cell differentiation and cancer development.^{124,125} Mutations influencing the structure of LAMP2 lead to the LSD Danon disease, which affects less mental, but rather muscle and cardiac functions and is discussed to be a glycogen storage disease.¹²⁶ In patients with sporadic PD, the level of LAMP2 was decreased in brain tissue¹²³ and leukocytes.¹²⁷ Moreover, PD patients with mutations in leucine-rich repeat kinase 2 (LRRK2) – a well described risk factor for sporadic PD – showed decreased LAMP2 concentrations in CSF.¹²⁸ However, the correlation with PD duration and progression, and the exact correlations with homo- or heterozygote variants remain unclear.¹²⁹ As autophagy is an essential step in degradation of aSyn, dysfunctional LAMP2 might be indirectly causative for PD via impaired autophagosomal processes.^{130,131}

Membrane proteins often serve as importers or exporters of specific substrates. A reduced number or a lack of functional exporters can cause accumulation of (toxic) molecules. The P₅-type ATPase 13A2 (ATP13A2, also known as PARK9) exports zinc and manganese ions^{132,133} as well as polyamines.^{134,135} Accumulation of manganese leads to the permeabilization of the lysosomal membrane¹³⁶ and enhances aSyn aggregation.¹³² Polyamines regulate levels of reactive oxygen species, not only affecting lysosomal but also mitochondrial stress response and toxicity, which furthermore leads to cell death and neurodegeneration.^{134,135} Mutations of the *ATP13A2* gene lead to EOPD and/or Kufor-Rabek syndrome

(KRS).^{137–139} As KRS has a juvenile onset and shows various neurodegenerative symptoms such as pyramidal dysfunction, spasticity, dementia, and parkinsonism with fast progression,^{140,141} it is often declared as a specific form of EOPD. Yeast models, animal models, and neuron cultures initially showed that aSyn aggregation and neurodegeneration can be rescued by expression of wild-type ATP13A2 or its homologues, as build-up of toxic manganese levels is inhibited.¹³² Mutations in *ATP13A2* cause different types of dysfunctions in neuronal cells: while homozygous mutations (including KRS mutations) often prevent the correct lysosomal localisation of ATP13A2 and/or lead to its destabilization, heterozygous mutations are more likely to affect the ATPase activity, thus the functionality of the exporter.¹⁴² Fibroblasts and dopaminergic neurons from ATP13A2-deficient PD patients showed altered lysosomal function, resulting in early cell death, as alteration of the lysosomal pH and permeabilization of the membrane occurred. Maturation and function of lysosomal cathepsins was impaired, thus aSyn aggregation increased in parallel with decreased depletion of accumulated toxic substrates.^{137,143} Furthermore, the role of zinc toxicity is of interest, as an excess of zinc causes glycolytic dysfunctions, which could be restored by ATP13A2 overexpression and pyruvate supplementation.¹⁴⁴ This example demonstrates the importance of detailed knowledge about a disease's pathomechanism, as possible treatments might depend on the underlying mutations and dysfunctions.

Unlike other lysosomal membrane proteins, the proton-activated transmembrane protein 175 (TMEM175) is expressed in most tissues and highly conserved in all domains of life, serving as an essential potassium channel. It maintains lysosomal integrity by regulating both pH and membrane potential, and membrane interaction or fusion.¹⁴⁵ The essential function as an antagonist (i.e. proton efflux pump) towards vacuolar-type ATPases, which provide proton influx into the lysosomal cavity, was recently described.¹⁴⁶ By altering the lysosomal pH, enzymes can lose their catalytic activity or stability (see introduction). Mutations or co-factors enhancing the activity of TMEM175 go along with increased lysosomal pH, while reduced activity leads to acidification of the lysosomal lumen.¹⁴⁶ Studies of (neuronal) cell lines with TMEM175 deficiency revealed a decrease of 20 to 35% in activity of the lysosomal cathepsins and GCase. Knockout of TMEM175 also reduced fusion of autophagosomes with lysosomes, thus accumulation of autophagosomal substrates occurred due to insufficient uptake and degradation by lysosomes. Both effects cause accumulation of aSyn, increasing the risk of developing PD.¹⁴⁷ In rat primary hippocampus neurons bearing the p.M393T mutation, dysregulated pH levels, reduced autophagy, and accumulation of aSyn was observed.¹⁴⁸

Impaired localization of TMEM175 remains to be discussed controversially. Apoptosis regulators of the Bcl-2 family can directly bind and inhibit TMEM175, although the mechanism is not fully understood. As Bcl-2 itself can be bound and inhibited, TMEM175 activity reduction or enhancement can be controlled and might serve as a therapeutic target.¹⁴⁹

Vesicular transport of lysosomal enzymes and lysosomal segregation

Besides full enzymatic activity or functionality, all lysosomal proteins have to be transported at the right quantity in order to fulfil their proper function in the lysosome. As described above, most lysosomal enzymes utilize the mannose 6-phosphate transport system to reach the lysosome. However, for some individual enzymes specialized transport proteins and signals developed, of which the lysosomal integral membrane protein type-2 (LIMP-2, coded by the *SCARB2* gene) is the best-studied example. It forms a complex with the GD- and PD-associated enzyme GCase in the ER. This complex is then trafficked through the Golgi apparatus and into the lysosome.^{20,21} While GCase transport appears to be the primary function of LIMP-2, recent studies have suggested an involvement of LIMP-2 in cholesterol transport.¹⁵⁰ A lack or dysfunction of LIMP-2 causes action myoclonus-renal failure syndrome. Besides renal failure, patients frequently report tremor and other neurological symptoms.¹⁵¹ Interestingly, *SCARB2* is also a risk gene for PD, most likely acting via the GCase-aSyn axis.¹⁵² Recent studies suggest that LIMP-2 furthermore acts as an activator and/or chaperon for GCase, as enhanced GCase activity could be measured *in vitro* in the presence of LIMP-2 compared to GCase alone.^{153,154} As a side note, LIMP-2 can also be expressed at the cell surface and thus also functions as the receptor for coxsackie virus, causing hand, foot, and mouth disease.¹⁵⁵

Another interesting, yet barely understood protein is the lysosomal cobalamin transport escort protein (LMBRD1). Beside cobalamin export,¹⁵⁶ it forms a complex with the ATP-binding cassette protein ABCD4.¹⁵⁷ Mutations in ABCD4 cause symptoms affecting different organs, developmental delay, and early death. Mutations in LMBRD1 and ABCD4, respectively, cause vitamin B₁₂ deficiency and accumulation of cobalamin in the lysosomes, leading to a lack of homocysteine and methionine^{157,158} and causing the Cobalamin F-type disease.¹⁵⁶ It was hypothesized that both trafficking from the ER to lysosomes and proper functionality is only possible after complex formation of both proteins. In parallel, LMBRD1 fulfils a role in endocytosis by co-localization and co-internalization of the insulin receptor in the plasma membrane, thus being an essential part of intracellular signalling and of main-

taining homeostasis of glycolipid metabolism.¹⁵⁹ Although no direct genetic link has been drawn to PD so far, symptoms often affect neuronal tissues, resulting in blindness, neurological defects, cerebral atrophy, and seizures. Whether or not LMDRB1 is linked to PD pathology has to be determined in future research.

Next to mention in lysosomal trafficking are cyclin-G-associated kinases (GAKs), which do not only serve as serine/threonine kinases, but also interact with clathrin adaptors, serving as co-chaperones for heat shock protein 70. Clathrin is a coating protein essential for lysosomal formation, endocytosis, and exocytosis. A knockdown of GAK can partially inhibit the endocytosis of clathrin-coated vesicles, leading to impaired vesicular transport, signal transmission, or protein recycling.^{160,161} Furthermore, the C-terminal domain binds a pre-form of CTSD (see chapter “ α -Synuclein degradation by lysosomal cathepsins”) for direct lysosomal sorting.¹⁶² GAK can increase PD risk, as a lack of substrates (kinase activity), accumulation of aSyn (binding partner for CTSD), and inhibited autophagy (involvement of clathrin uncoating) can lead to PD-associated symptoms in neuronal and non-neuronal cell models independently of each other.¹⁶³

Ras-related proteins (Rab family; reviewed in¹⁶⁴) are membrane proteins with important roles in vesicle transport and fusion, neuronal maturation, and many other functions. Alteration in almost any member of this family can lead to severe symptoms, e.g., four of them are associated with neuronal dysfunctions, and Rab39B has been correlated with intellectual disabilities and is classified as an EOPD risk factor, as certain mutations cause destabilization and loss of function in RAB39B proteins.^{165,166} Possible consequences are aSyn aggregation and Lewy body formation.¹⁶⁴

Characterization of lysosomal enzymes – activity vs. protein levels

Quantification of enzyme activity in patient-derived samples like fibroblasts or blood-derived samples (plasma, leukocytes, dry blood spots) is a widely used tool for screening and diagnosis of LSDs. Activity measurements often rely on fluorometric or colorimetric activity assays using an acidic reaction buffer and a synthetic substrate.¹⁶⁷ For many lysosomal enzymes, 4-methylumbelliferyl (4-MU) conjugates have been used successfully as substrates (including the following enzymes from Table 1: GCase, GALC, glycosaminoglycan α -L-iduronohydrolase, iduronate 2-sulfatase, ARSA/B, N-acetylgalactosamine-6-sulfatase, α -L-fucosidase, α -galactosidase A, HEXA/B, neuraminidase 1, sulfamidase, acid sphingomyelinase).^{168–181} For lysosomal cathepsins, fluorogenic peptides with enzyme specific cleavage sites can be used and fluorometric activity assay

kits are commercially available. In an alternative approach, turnover on non-fluorogenic enzyme substrates can be measured by using tandem mass spectrometry, allowing increased detection range and multiplex analyses.^{182,183} For some enzymes (GALC, acid sphingomyelinase, lysosomal acid lipase), the missing specificity of the 4-MU substrate forces the use of radiolabelled native substrates, which is used in diagnostics for higher specificity.¹⁸⁴

A downside of activity measurement in whole-cell lysates and peripheral samples is the disregard for cellular localisation of the target enzyme. Since maturation and activity of lysosomal enzymes is mediated by acidic pH and other lysosomal enzymes,¹⁸⁵ lysis of whole cells and addition of an acidic reaction buffer in an activity assay might lead to activation of enzymes that have not undergone full maturation and lysosomal delivery. Therefore, these assays overlook possible changes in lysosomal trafficking, which is regarded as one of the major components of PD pathology. This disturbance of lysosomal trafficking and disruption of lysosomal functions in midbrain neurons by aSyn was shown in a study by Mazzulli *et al.*,¹⁸⁶ which encompassed a variety of experimental approaches to investigate lysosomal functions and trafficking. A pulse-chase experiment using radiolabelled lysine was conducted to measure protein degradation, allowing quantification of overall lysosomal function in a broadband approach.¹⁸⁶ In addition, lysosome-specific live cell activity assays of human midbrain dopamine neuronsshowed significantly reduced activity of CTSD, GCase, β -galactosidase, hexosaminidase, and sulfatase in PD and/or GD patients (at least half of the controls' activity or less). Noteworthy, while lysosome-specific GCase activity was significantly reduced under high aSyn conditions, total cellular GCase activity was unaffected, demonstrating the discrepancy between whole-cell and lysosome-specific activity. Finally, western blot analyses and immunofluorescence staining confirmed the disturbance in lysosomal transport of hydrolases in presence of pathological aSyn conformers.¹⁸⁶ This study is a good example on how to assess lysosomal enzyme function while considering the crucial trafficking component. A comprehensive methodological guide to studying trafficking and activity of lysosomal enzymes in human iPSC-derived neuronal models has also been published by Cuddy and Mazzulli and we encourage resorting to this for guidance when addressing similar scientific questions.¹⁸⁷

It is important to note that enzyme activity and protein levels are not interchangeable readouts, especially in the context of disease-associated genetic variants that affect an enzyme's expression and/or activity. Thus, both readouts should be considered to obtain a full picture. In general, quantification of lysosomal protein levels can be done, for example, via western blot

analysis,⁵⁷ or mass spectrometry.⁶⁰ In PD/LSD research, quantification of misfolded proteins, protein oligomers, and protein aggregates also play key roles, which might require more complex immunoassays such as different types of ELISA, fluorescence microscopy or spectroscopy, microbeads, radiolabelling, or others.¹⁸⁸ In addition to information on protein abundance, these methods can also provide qualitative information on protein processing and maturation through analysis of maturation-specific band patterns,^{186,189} or by characteristic peptide fingerprints.⁶⁰ Enrichment or purification of lysosomes from cell homogenate via subcellular fractionation⁷⁶ or immunoprecipitation¹⁹⁰ can be utilized prior to western blotting to separate lysosomal and non-lysosomal species, allowing more detailed analysis of intracellular trafficking. Additional information about maturation state can also be obtained by EndoH and PNGaseF digest of western blot samples. These enzymes selectively cleave pre-Golgi and post-Golgi N-linked glycans, respectively, providing additional insight into the maturation state and subcellular distribution of the target enzyme.¹⁸⁷

Lysosomal enzymes as biomarkers and diagnostic tools in PD

To date, the clinical diagnosis of PD is based on the presence of characteristic neurological symptoms.¹⁹¹ While a molecular diagnosis was previously only possible *post mortem* by examination of the brain, recent advances in the understanding of molecular mechanisms have sparked the search for diagnostic markers in biosamples of PD patients. So far, most of these biomarkers focus on aSyn as the hallmark molecule of PD. However, as touched upon in the previous chapter, activity measurements of lysosomal enzymes are well-established diagnostic tools for LSDs, and with the increasing evidence of connections between LSDs and PD, changes in lysosomal enzymes could potentially function as a biomarker for PD diagnosis and/or prognosis.

Due to the tight involvement of GCase in PD pathology, studies have explored GCase activity in biofluids as a potential biomarker for PD. Several studies analysed CSF and found decreased GCase activity in PD patients.^{116,192–194} Importantly, while GCase activity was most decreased in GBA-PD patients and correlated with the severity of the *GBA1* mutation,¹⁹³ even PD patients without *GBA1* mutations exhibited reduced CSF GCase activity.¹⁹⁴ In addition to GCase activity, a reduced activity of α - and β -mannosidase¹⁹² and CTSD¹⁹⁴ was reported. Changes in the activity of β -HEX varied between the studies, with two studies showing a slight, statistically non-significant decrease^{192,194} while another study reported an increase in activity.¹¹⁶ There is, however, a conflicting study by van Dijk *et al.* that found unchanged GCase, β -HEX, β -mannosidase

and CTSD, but increased β -galactosidase and cathepsin E activity.⁵⁹ This study also raises the important point that CSF samples are generally obtained for diagnostic purposes from individuals with neurological symptoms. Therefore, while the control samples do not stem from patients diagnosed with PD, they do not necessarily reflect “healthy” controls. Nonetheless, it appears that there are alterations in lysosomal enzymes, namely GCase, in CSF of PD patients. The studies mentioned here suggest that while measurement of the activity of a single lysosomal enzyme in CSF samples might not be a sufficiently accurate biomarker for PD, looking at a set of lysosomal enzymes could improve specificity of a molecular PD diagnosis.

Besides the analysis of CSF samples, great efforts have been made to identify PD biomarkers in biofluids that are less invasive and easier to obtain, such as urine samples or peripheral blood. Proteomics analysis of urine samples by Virreira Winter *et al.* found a dysregulation of lysosomal proteins (e.g., CTSB, HEXA, HEXB, GALC), but only in PD patients harbouring the *LRRK2* p.G2019S mutation.¹⁹⁵ In peripheral blood, Alcalay *et al.* showed that GCase activity in dried blood spots of GBA-PD, LRRK2-PD, and sporadic PD patients was decreased compared to controls. Furthermore, GCase activity correlated with disease duration, suggesting that lower GCase activity led to faster disease progression.¹⁹⁶

The most exciting advance in recent biomarker research has been the exploitation of extracellular vesicles (EVs). These vesicles are secreted by cells for communication, cargo transport or to get rid of cellular waste (see¹⁹⁷ for a comprehensive review on EVs). Thus, they carry cell type-specific cargo and surface markers. Importantly, EVs can pass the blood–brain barrier (BBB), allowing the isolation of brain-derived EVs from peripheral blood. While the idea of looking at brain-derived EVs in blood originated in the AD field,¹⁹⁸ the immense potential of this new approach for PD has been demonstrated by a study of Kluge *et al.*, in which brain-derived pathological aSyn was detected in blood samples of PD patients via isolation of neuronal exosomes.¹⁹⁹ It remains to be seen whether neuronal exosomes can also shed more light on the dysregulation of lysosomal enzymes in the living brains of PD patients and whether detecting these changes can serve as a reliable, consistent diagnostic or prognostic biomarker standard for PD.

Lysosomal dysfunction and aSyn in LSD and PD models

To study neurodegenerative diseases, animal models often serve as platform to reflect human pathology as close as possible. One of the earliest PD models was a 6-Hydroxydopamine (6-OHDA, a dopamine analogue) induced neurotoxin rat

model, which leads to PD development after metabolization of the 6-OHDA in neurons and consequent mitochondrial dysfunction.²⁰⁰ Also, other toxin-induced models are in use, provoking accumulation of reactive oxygen species for example by plant or fungi toxins, and further causing aSyn inclusions.²⁰¹ 1-methyl-4-phenyl-1,2,5,6-tetrahydropyridine (MPTP)-induced models which reflect most of the PD hallmarks except Lewy body formation²⁰² currently serve as a gold standard in (toxin-induced) animal models for PD. Instead of toxins, viral vectors are also used, for instance to initiate over-expression of aSyn, leading to significant loss of dopaminergic neurons.^{203–205} However, both systems often fail to reflect the broad spectrum of PD pathology and malfunction-consequence-circles caused by impaired protein functionalities. Genetic models can target specific proteins and pathways, and many examples and models exist especially for aSyn aggregation in the brain, either by altering the levels of aSyn degrading enzymes like cathepsins⁵⁴ or by the expression of mutated aSyn genes like the variants p.A30P and p.A53T.²⁰⁶ LRRK2 overexpression in mouse models could provoke different PD-correlated effects, such as altered motor and behavioural performance, decreased proliferation and survival rates of cells, or reduced neurotransmission in dopaminergic neurons,^{207–210} which helps in understanding and treating PD forms provoked exclusively by LRRK2 mutations. Since LSDs are normally caused by more specific cellular malfunctions, i.e. single proteins, it is easier to establish suitable animal models for these than for multifactorial disorders like PD. Two prominent examples are the previously mentioned CTSD-deficient NCL10 mouse⁵⁴ and a mouse model for GD carrying distinct *GBA1* mutations²¹¹ or gene deficiency.²¹²

To date, no *in vivo* model can fully recapitulate the complexity of PD. Mechanisms in neuronal cell function such as the dopamine metabolism²¹³ as well as aSyn aggregation^{214,215} differ between humans and rodents. For instance, murine neurons often lack hallmark symptoms like Lewy body-like fibrils²¹⁶ or behaviour and motor phenotypes.²¹⁷ Toxin-induced models can cause adverse or stronger effects than aimed for as they might affect not only the target protein or cellular component. When establishing aSyn induced animal models, the promoter has to be chosen carefully and differentiated between human and rodent aSyn expression.

Thus, use of animal models has to be evaluated with caution, yet they still provide a very valuable tool to study single, straight defined biochemical or molecular questions about mechanisms in LSD or PD pathology and correlations to lysosomal (dys)functions.^{218,219}

To bridge the knowledge gaps of human cellular processes in PD, new methods emerged and improved over the last decades. A powerful approach is the use of induced pluripotent stem

cells (iPSCs), which allows PD patient-derived cells to be re-programmed into dopaminergic neurons,²²⁰ enabling further characterisation of lysosomal function in living neuronal cells instead of patient derived *post mortem* brain tissue and/or CSF samples. Dopaminergic neurons can be maintained *in vitro* over months and *in vivo* survival was proven in mice, rats, and monkeys.²²⁰ This approach also allows a more detailed analysis of cellular/lysosomal function in neuronal cells both in the presence and the absence of aSyn. Notably, a wide variety of genetic backgrounds can be analysed, ranging from high impact mutations in the aSyn gene *SNCA*²²¹ to genetic risk factors such as variants of *GBA1*²²² and *LRRK2*.²²³ Besides iPSCs equipped with down- or dysregulated proteins, multiplication of certain (risk) genes is another tool for studying the consequences of altered protein levels and function. As an example, both the duplication and triplication of the *SNCA* locus was found to increase activity of caspases, promote protein aggregation, oxidative stress, and cell death.^{221,224} The analysis of GD and PD patient derived iPSCs heterozygous for *GBA1* mutations helped to prove the accumulation of glucosylceramide as a consequence of the gene variants, which was then linked to aSyn aggregation, impaired lysosomal or autophagosomal function, and dysregulated calcium homeostasis.²²² Controlling cell differentiation gets more and more precise and robust, e.g. by exposure to neurotoxins like MPP⁺²²⁵ or retinoic acid,²²⁶ so that symptoms of PD can be provoked in iPSCs. Several iPSCs models have been established as a good base for proteomics, transcriptomics, and metabolomics.²²⁷

Most intriguingly, iPSC technology allows individualized, per-patient characterization of neuronal pathology, which is particularly important for the understanding of the molecular mechanisms underlying those EOPD cases, which lack common PD risk factors.²²⁸ On one hand, this enables the exploration of shared pathomechanisms of patients with different genetic background, and also offers the potential of paving the way towards more personalized treatment and diagnostic strategies in the future.^{229,230} On the other hand, iPSC models highly depend on the genetic background of each line and need a clear association to a genetic risk factor to interpret changes in cell or organelle morphology, expression levels, or transduction. Furthermore, for PD studies the use of mature dopaminergic cells is crucial, yet functional and cellular properties can drastically change over time,²³¹ and two-dimensional cell cultures can neither fully reflect functional three-dimensional brain tissue nor organic implements like the BBB.²³² Attempts of using co-cultures with different cell types to mimic the physiological conditions in PD progression were done, e.g. demonstrating that astrocytes are crucial for regulation and maintenance of neurons, neuronal networks, and neurotransmit-

ters.^{233–235} Different methods of using iPSCs to study LSDs and PD have been reviewed in.²³⁶

Therapeutic potential of lysosomes and lysosomal enzymes

As a result of intensive research in the field of both LSDs and neurodegenerative diseases, increasing number of genetic alterations and changes in the molecular level have been identified, which suggest similar pathomechanisms underlying the two disease groups. However, important differences between PD and LSDs should not be overlooked: LSDs frequently present severe systemic symptoms affecting organs other than the CNS. In light of this, therapeutic approaches often target treating and/or alleviating symptoms that are caused by the dysfunction of peripheral organs. Another significant difference between LSDs and PD is, while the former ones are generally results of gene mutations manifesting frequently early in life, the latter has a more diverse aetiology and it is more associated with advanced ages. Nonetheless, if a therapeutic approach tackles a molecular mechanism in LSDs which decreases ongoing neurodegeneration, it could hold great potential for the treatment of neurodegenerative disorders as well, including PD.

Still, an obstacle that needs to be overcome in the treatment of PD and neurodegenerative disorders is the BBB. Worth noting here is that nowadays, a view of PD originating from the gut is also gaining attention.²³⁷ Clarification of the involvement of the gastrointestinal track and enteric nervous system in PD could give more insight into the origin of the disease and lead to the identification and development of new therapies targeting non-brain regions. In the following description of possible therapeutic approaches targeting lysosomal enzymes/functions, we focus on PD while only mentioning selected types of established LSD treatments. For a more detailed view, we call the readers' attention to recent reviews on the topic.^{238,239}

The interventions used or attempted for the treatment of LSDs can be grouped into four methods: i) replacing/substituting faulty enzymes with functioning enzymes using enzyme replacement therapy (ERT) and gene therapy approaches; ii) modification of enzyme activity with the use of small molecules targeting the enzyme's active site or as allosteric regulators, i.e. pharmacological chaperon therapy (PCT); iii) adjustment of substrate levels to reduce an enzyme's activity, i.e. substrate reduction therapy (SRT); and adjuvant therapies (see Figure 2). In theory, either of these ways can be a reasonable approach for PD treatment as well. An overview of novel as well as potential therapeutic strategies for PD targeting the lysosome can be found in the following.

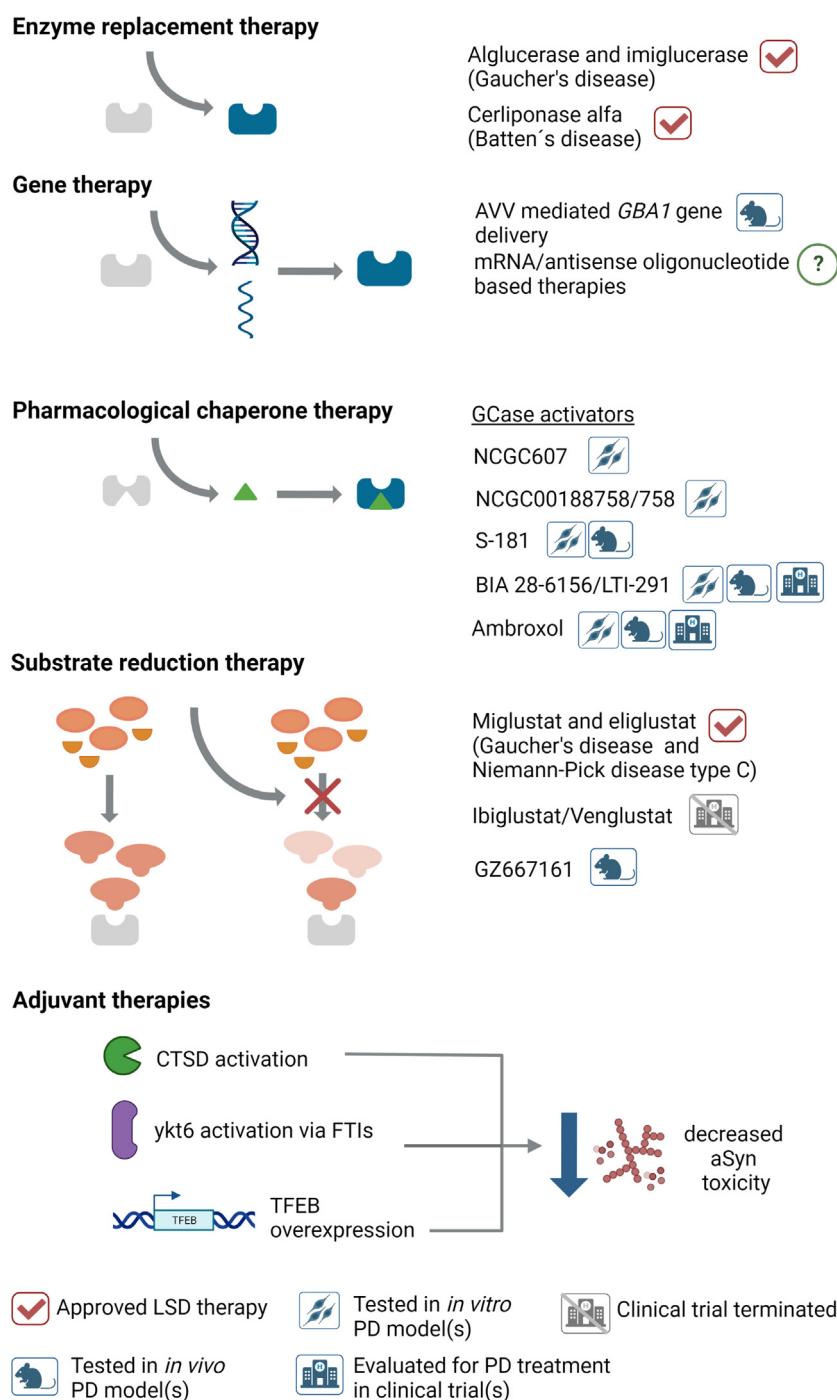


Figure 2. The increasing evidence of shared pathomechanisms between LSDs and PD suggest that therapeutic interventions established or currently under development for the treatment of LSDs hold the potential of being suitable for the treatment of PD as well. These interventions can be grouped as methods aiming and replacing/substituting the faulty or missing enzyme (enzyme replacement and gene therapy approaches), enhancing residual enzyme activity with the use of small molecular active site or allosteric regulators (pharmacological chaperone therapy), adjusting the substrate levels of enzymes with reduced activity (substrate reduction therapy), and adjuvant therapies. This figure gives an overview of these approaches, with a special focus on PD, and shows a selection, but not a complete list of compounds that have been or are currently evaluated in pre-clinical studies and/or clinical trials. The icons next to the exemplary compound names indicate whether the drug has been evaluated in *in vitro* (tissue culture icon) or *in vivo* (mouse icon) models of PD, or entered clinical trial(s) (hospital icon). The red checkmark icon indicates if a compound has been approved for the treatment of an LSD. Abbreviations: AVV, adenovirus vector; *GBA*, β -glucocerebrosidase (gene); GCase, β -glucocerebrosidase (protein); CTSD, cathepsin D; TFEB, transcription factor EB; FTI, farnesyl transferase inhibitor; aSyn, α -Synuclein. (Figure created with [BioRender.com](https://www.biorender.com/).)

Enzyme replacement therapies (ERTs)

The first ERT was approved for the treatment of GD in 1991.^{240,241} Since that time, ERTs for more than ten LSDs have been adopted or are currently under clinical trials.^{242–244} This therapy type is considered as a standard treatment for multiple LSDs; however, it has certain limitations. These are high cost of recombinant enzyme production, risk of infection due to repeated application, and the risk of eliciting immune responses.³⁶ A further limitation of such therapies is that enzymes – produced regularly by the use of recombinant DNA technology, frequently in modified forms – are large molecules unable to cross the BBB, thus their application is often limited for the treatment of non-neurological symptoms.³⁶

A possible approach for overcoming the problem of BBB penetrance is to deliver the recombinant enzyme directly into the CSF. Possible routes for this are the intracerebroventricular and intrathecal enzyme delivery.^{36,239} Direct enzyme delivery has been recently approved for the treatment of neuronal ceroid lipofuscinosis type 2 (CLN2).²³⁹ CLN2 is a form of Batten's disease, caused by the deficiency of the lysosomal enzyme tripeptidyl peptidase 1. Intraventricular application of a recombinant tripeptidyl peptidase 1 proenzyme, cerliponase alfa (Brineura®), resulted in diminished decline in motor and language skills in children suffering from CLN2.²⁴⁵ The approval of cerliponase alfa for the treatment of CLN2 in 2017²³⁹ demonstrates the potential of this therapeutic approach, and gives hope for the success of applying direct enzyme delivery into the CNS for the treatment of other neurological disorders as well. However, intracerebroventricular and intrathecal enzyme delivery is not without risk. Adverse effects can include malfunctioning of the intraventricular reservoir and leakage, and even life-threatening infections.²⁴⁵ Thus, regular medical control and high level of patient compliance are essential. As an alternative approach, an ongoing clinical trial [NCT04370665] is exploring deliberate disturbance of the BBB as a way to deliver ERT to the brain.

Gene therapy

Another method to reconstruct a functional enzyme could be by establishing an endogenous source by the delivery of a copy of the gene coding the functional protein. Gene therapy methods involve both *in vivo* and *ex vivo* approaches: the former utilizes a viral vector to carry the transgene into the target cells, while in the case of *ex vivo* gene therapy patient derived cells are genetically modified *in vitro*, and the altered cells are then re-implanted into the patients.³⁶ Transfer of a transgene/genetic modifier to the CNS poses a great challenge. Similar to ERTs, a possible way to overcome the limited access of the BBB, is to conduct gene transfer

directly to the brain which has been applied in clinical trials for the treatment of various LSDs, such as mucopolysaccharidosis, metachromatic leukodystrophy, NCL, and GD (for a detailed review see.²⁴⁶

Results of pre-clinical studies suggest that gene therapy targeting restoration of impaired/defective lysosomal enzyme function has its potential in the treatment of PD as well. Intracerebral adenovirus vector (AVV) mediated *GBA1* gene delivery in *in vivo* murine models (aSyn overexpressing mice and a rat model in which selective dopaminergic neuron loss was induced by AVV mediated delivery of the p.A53T mutant aSyn) ameliorated aSyn accumulation in the *striatum* and *substantia nigra* of the animals.²⁴⁷ In a more recent study using the A53T-SCNA transgenic mouse model, Morabito and colleagues demonstrated that systematically administered AVV carrying the *GBA1* transgene resulted in increased GCase expression in the CNS, which was accompanied by reduction of aSyn pathology.²⁴⁸

Among the limitations of gene therapy approaches is that the knowledge of genetic background of the targeted disease is prerequisite in order to correct it. Since LSDs are usually monogenic diseases, they are promising targets for such interventions.²⁴⁹ In contrast, PD is a multifactorial disease and pathogenic/disease causing mutation of a single gene can be identified only in a small percentage of the cases. In order for a gene therapy approach to be viable, understanding the exact pathomechanisms and genetic variations on an individual basis is essential.

Pharmacological chaperone therapies (PCTs)

An innovative approach for correcting enzyme deficiency is to use so-called pharmacological chaperones. Endogenous chaperones play a crucial role in stabilizing protein conformation, preventing protein misfolding and aggregation, thus maintaining protein function. Incorrectly folded and/or unstable proteins are detected by the quality control systems of the cells and are marked for degradation. Missense mutations in lysosomal enzymes can impact the structure of the protein, resulting in misfolding and consequent impairment in its activity and/or trafficking to the lysosome.^{250,251} Application of pharmacological chaperones can enable the stabilization of these proteins, thus preventing their premature degradation and facilitating their transport to the lysosomes and biological activity.²⁵⁰

A great advantage of PCTs is that they can utilize small molecule drugs that are applicable orally and are capable of crossing the BBB.³⁶ However, since this approach targets the endogenous enzyme, it can be only applied in patients with clearly defined mutations.²⁵² A further aspect that needs to be considered is the possible interaction of a pharmacological chaperone with the active site of a protein, thereby serving as a competitive inhibitor. In order

to avoid this, development of compounds binding outside of the catalytic domains is favourable.^{36,253}

The high prevalence of *GBA1* mutations among PD patients puts GCase in the spotlight of PD research. Various molecules have emerged as promising candidates for modifying GCase activity in PD, however, most of them have only been evaluated in pre-clinical studies so far. Among these compounds are new developments, such as NCGC00188758 (also known as 758),^{254,270} NCGC607,²⁵⁵ and S-181,²⁵⁶ all small-molecule GCase activators. Studies involving iPSC models showed that all three compounds successfully enhanced GCase activity and decreased aSyn pathology.

Recently, a BBB penetrant allosteric GCase modulator, BIA 28-6156/LTI-291 has emerged as a promising candidate for the treatment of GBA-related PD. Preclinical studies showed that the compound increased GCase activity and decreased glucosylceramide levels *in vitro* and *in vivo*.³³⁷ BIA 28-6156/LTI-291 was well tolerated and caused no serious adverse effects based on the results of phase I clinical trials involving healthy volunteers and PD patients (Netherlands Trial Register NTR6598, NTR6705,³³⁷ and NL6574).³³⁸ According to a press release of BIAL Biotech in 2020, BIA 28-6156/LTI-291 has entered a phase II clinical trial in 2021.³³⁹

Another well-known and commonly used GCase-modulating drug is the expectorant ambroxol. Pre-clinical studies have been utilizing various *in vitro* models, i.e., GBA-mutant PD patients derived fibroblasts,²⁵⁷ GD fibroblasts,²⁵⁸ GD and GBA-PD patient derived fibroblasts,^{62,259} and mouse cortical neurons.²⁶⁰ Additionally, *in vivo* models, i.e., human aSyn overexpressing and heterozygous murine L444P-GBA expressing mice,²⁶¹ human WT, N370S-GBA, and L444P-GBA expressing *Drosophila*,²⁵⁹ and PD, GD, and GBA-PD patient derived human macrophages²⁶² showed that ambroxol increases both protein expression and activity of GCase. These effects were accompanied by a reduction of aSyn and phosphorylated aSyn levels and a decrease in the loss of dopaminergic neurons.²⁶³ These promising findings led to a phase II clinical trial conducted between January of 2017 and April of 2018 ([Clinicaltrials.gov](https://clinicaltrials.gov/ct2/show/study?term=NCT02941822) identifier: NCT02941822). The findings indicate that ambroxol is brain penetrant and results in increased aSyn and GCase protein levels in the CSF. Furthermore, the treatment was well tolerated and no serious adverse effects were reported.²⁶⁴

Substrate reduction therapies (SRTs)

A further mean by which excess accumulation of toxic metabolites can be prevented is via the inhibition of a substrate's production.³⁶ This is the goal of substrate reduction therapies (SRTs). SRTs have various advantages over ERTs. First, they are more likely to be BBB penetrant, enabling better

treatment of CNS-related symptoms. Second, oral availability and non-immunogenicity oppose only small impact on quality of life and require less patient compliance compared to ERTs.^{36,265}

Examples for SRTs are drugs approved for the treatment of GD and Niemann–Pick type C disease: the imino sugar miglustat (Zavesca®) and the ceramide analogue eliglustat (Cerdelga).^{266,267} Miglustat and eliglustat are inhibitors of glucosylceramide synthase, an enzyme responsible for the formation of glucocerebroside (glucosylceramide).²⁶⁷ But even though miglustat is capable of crossing the BBB,^{267,268} it did not improve the neurological symptoms of patients suffering from GD significantly.²⁶⁹

Results of Zunke *et al.*²⁷⁰ suggest that decrease of glucosylceramide levels is also a promising approach against aSyn pathology. *In vitro* studies showed that the interaction of aSyn and glucosylceramide leads to a structural conversion of the protein into a pathogenic conformation. These high-molecular weight aSyn species were found to be capable of recruiting and converting soluble forms of the protein into insoluble aggregates, and induced cellular degeneration. Decreasing glucosylceramide levels either by the glucosylceramide synthase inhibitor eliglustat or via a small molecule GCase activator, NCGC00188758 or 758, restored the physiological conformers of the protein and decreased aSyn pathology.²⁷⁰

Ibiclustat (also named Venglustat or Genz-682452) is another brain-penetrant SRT compound developed with the aim of treating CNS-related symptoms of LSDs. It was also evaluated as a potential treatment for PD, acting via inhibition of glucosylceramide synthase to lower glucosylceramide levels and thus slow aSyn aggregation.²⁶⁷ In a phase II clinical trial involving early-stage GBA-PD patients (MOVES-PD, NCT02906020) ibiclustat treatment resulted in diminished plasma and CSF glucosylceramide levels. However, the study had to be terminated due to adverse clinical outcomes.²⁶³ Nonetheless, the brain penetrability of SRTs gives hope and warrants for the involvement of these compounds in further research on PD treatments. Although so far it has been only tested in pre-clinical experiments, treatment of GbaD409V/D409V and A53T–SNCA transgenic mice (mouse models of GD and PD/synucleinopathy, respectively) with GZ667161 resulted in decreased glucosylceramide and glucosylsphingosine levels in the CNS, slowed hippocampal aSyn accumulation and ameliorated memory and cognitive deficits in the animals.²⁷¹

Adjuvant therapies

Adjuvant therapies, which target secondary pathways affected in the disease, hold the potential of improving life quality and slowing down disease progression of patients with a variety of pathomechanisms. Since aSyn

accumulation and the consequent cell loss are pathological hallmarks of PD, but not necessarily caused by direct *SNCA* mutations but rather side-effects of other dysfunctional proteins, any reduction of aSyn aggregate formation could be a viable therapeutic approach. CTSD is a promising target for preventing aSyn accumulation, since findings of studies using mice and *Caenorhabditis elegans* models showed that CTSD plays a pivotal role in its degradation. Deficiency of the enzyme was reported to be associated with aSyn accumulation, while increased expression of CTSD was found to have a protective effect against aSyn induced toxicity (for a detailed review see^{243,272}).

Ykt6, a soluble N-ethylmaleimide-sensitive-factor attachment protein receptor (SNARE) protein, has recently emerged as a potential candidate for intercepting aSyn related pathology. In their study, Cuddy *et al.*²⁷³ demonstrated that under metabolic stress, ykt6 moves into the membranes from the cytosol, thus promoting hydrolase trafficking and lysosomal activity. Blocking its farnesylation with the small molecule farnesyl transferase inhibitor LNK-754 increased ykt6 activity and enhanced trafficking of lysosomal GCase, and reduced aSyn pathology in PD patient iPSC-derived dopaminergic neurons and a DA^{SYN53} mouse model of the disease.

A further promising adjuvant therapeutic approach is the modulation of TFEB. As discussed earlier, TFEB is a master regulator of lysosomal function, and controls lysosomal biogenesis and autophagy. Up-regulation of TFEB has been shown to have beneficial effects in various *in vivo* and *in vitro* models of PD. Torra *et al.*²⁷⁴ demonstrated that AAV mediated overexpression of TFEB in *substantia nigra* dopaminergic neurons of the MPTP induced mouse model of PD activated pro-survival pathways, and prevented cell loss induced by the toxin. AAV mediated TFEB overexpression in rats simultaneously overexpressing human aSyn protected both dopaminergic neurons and striatal terminals, which was proposed to be due to the reduction of aSyn oligomer formation and an increase in processing of autophagosomes.²⁸ In a study using aSyn overexpressing human neuroglioma cells, both genetic overexpression and 2-hydroxypropyl- β -cyclodextrin induced pharmacological activation of TFEB decreased accumulation of aggregated aSyn via increasing the autophagic clearance of the protein.²⁷⁵

Perspective – Challenges in LSD and PD diagnosis and treatment

The development of (EO)PD linked to mutations leading to the malfunction of a single protein occurs only in a small percentage of patients. However, lysosomal transport, sorting, and the whole process of substrate recycling is a fragile

system with plenty of interaction partners and converging mechanisms. The analysis of different LSDs in the last years have taught us valuable lessons and revealed mechanistic insight into lysosomal dysfunction in PD.²⁷⁶ Nevertheless, there are still some major shortcomings that need to be solved to drive lysosomal research not only in the context of PD, but also in other neurodegenerative diseases.

Currently, PD is diagnosed after the first clinical symptoms have already occurred, however, by that time neurodegenerative processes have already led to the loss of about 70% of dopaminergic neurons.²⁷⁷ Thus, reliable and standardized clinical approaches are needed for the recognition of the disease at early stage. One approach could be the detection of aggregated aSyn from blood samples as demonstrated recently.¹⁹⁹ Another possibility would include the development and use of highly specific substrates against lysosomal enzymes for the quantification of altered enzymatic activity, e.g., in human derived fibroblasts or leukocytes. As around 75% of LSDs are due to defects of lysosomal hydrolases,²⁷⁸ increasing the number and specificity of substrates to evaluate enzymatic activity would be one goal over the next years. However, evaluation of the activity of individual enzymes can only give a glimpse into the complex degradation network in the lysosome. Delivery of lysosomal proteins as well as substrates to the lysosome is just as crucial for lysosomal function as the integrity of the enzymes, however, signals for binding and release of protein-protein or protein-substrate interactions are mostly unknown.²⁷⁹ Moreover, biochemical and biomechanical feedback loops within the lysosome can provoke more than one factor to increase neurodegeneration (see graphical abstract), e.g. aSyn aggregation can be enhanced by many other dysfunctional proteins, even without the *SNCA* gene being affected.²⁸⁰ Development of more complex, yet native-like cell systems and organoids are urgently needed to understand lysosomal dysfunction in the PD disease context. These assays will need to include genetic modifications of more than one risk factor, with specific read-out tools at the same time. A cross-correlation and -evaluation between genetic analyses (demonstrating the cause of a protein dysfunction) and activity assays (demonstrating the output or consequence of this dysfunction) or other analysing methods will help to elucidate networks and interactions and offer the possibility of identifying the key players in the pathomechanisms of lysosomal disorders.

The final goal should be to establish translational approaches and clinical trials to transfer the knowledge from *in vitro* and *in vivo* models to patients, since this could help to identify lysosomal dysfunction preferentially in the prodromal phase of the diseases. While various LSDs present symptoms of neurodegeneration and/or

parkinsonism, in some cases the genetic and/or molecular background that is common with PD has not been identified up to today. However, one might ask the question, should a disease be defined by its symptoms, or by the underlying pathomechanisms? As for those cases with a described shared pathomechanism of LSDs and PD, the realization of numerous common mechanisms could pose the question: could (at least some of) these storage disorders be considered as a prodromal phase of PD, or PD with onset at a young age? The Kufor-Rabek syndrome could serve as an example in this regard, since it is often considered as a form of EOPD, but also as an LSD.^{141,281} Additionally, some genetic variations can either lead to an LSD or PD, respectively, depending on their homo- or heterozygous presence as well as the impact of the individual mutation on protein function, as shown for *GBA1*⁶² and *GALC*.⁸⁹ Hence identifying clear differentiations or convergences of LSDs and PD is difficult and is out of the scope of this review. We believe, however, that further research focusing on unravelling the similarities between LSDs and PD is highly warranted, as it will offer valuable information from which both fields of LSDs and neurodegenerative diseases are going to benefit. This would not only hold the potential of preventing irreversible neurodegenerative harm, but also of identifying the best treatment option for every patient individually.

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CRedit authorship contribution statement

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DATA AVAILABILITY

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DECLARATION OF COMPETING INTEREST

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Abbreviations:

4-MU, 4-methylumbelliferyl; aSyn, α -synuclein; ATP, adenosine triphosphate; AD, Alzheimer disease; AVV, adenovirus vector; BBB, blood-brain barrier; CLN2, ceroid lipofuscinosis type 2; CNS, central nervous system; CSF, cerebrospinal fluid; CTSx, cathepsin x; ER, endoplasmic reticulum; EOPD, early onset Parkinson's disease; ERT, enzyme replacement therapy; GD, Gaucher's disease; GWAS, genome-wide association studies; iPSCs, induced pluripotent stem cells; MLD, Metachromatic leukodystrophy; MPTP, 1-methyl-4-phenyl-1,2,5,6-tetrahydropyridine; NCL, neuronal ceroid lipofuscinosis; KRS, Kufor-Rabek syndrome; LSD, lysosomal storage disease/disorder; PCT, pharmacological chaperone therapy; PD, Parkinson's disease; SRT, substrate reduction therapy

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